

FakeOut: Rental Scam Classifier

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Abstract

Rental listing fraud costs American renters tens of millions of dollars annually, yet no public dataset of verified scam labels exists for long-term US rentals, making supervised fraud detection impossible without first solving a labeling problem. We investigate whether programmatic weak supervision can substitute for hand-labeled ground truth in this setting, using the UCI Apartments for Rent dataset (99,467 listings). We engineer price anomaly features against the Zillow Observed Rent Index (ZORI) via spatial k-NN matching, extract effective prices from aggregator listing bodies to correct a systematic floor-pricing bug, and implement 17 labeling functions across four families in a Snorkel pipeline. To evaluate the resulting labels, we hand-label 165 stratified listings across six targeted sampling rounds. The Snorkel LabelModel achieves AUROC 0.45 against ground truth, slightly below random, and a downstream gradient-boosted model trained on the weak labels performs worse still at AUROC 0.33, amplifying rather than correcting the label model's errors. By contrast, an L1 logistic regression trained via leave-one-out cross-validation on the 165 hand labels achieves AUROC 0.82. We show that this failure is structural: the corpus is dominated by legitimate aggregator listings whose below-market floor prices trigger scam-leaning labeling functions, poisoning the weak supervision signal. Our results demonstrate that biased labeling functions do not average out, they compound, and that a small number of carefully sampled hand labels can decisively outperform a large weakly-labeled training set when the underlying data distribution violates the diversity assumptions of data programming.

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1 Introduction

The digital transformation of the real estate market has streamlined the search process for renters but has simultaneously introduced significant systemic risks through fraudulent listings. These fake listings range from sophisticated phishing attempts to bait-and-switch advertisements, causing substantial financial loss and eroding trust in online marketplaces. Detecting these anomalies at scale is a non-trivial task due to the high variance in rental pricing across different micro-markets and the strategic mimicry used by bad actors to appear legitimate.

In this paper, we present FakeOut, a machine learning pipeline designed to identify fraudulent rental listings within the UCI Apartment Listings dataset ($\sim 99,000$ observations). The core technical challenge addressed in this research is the *labeling bottleneck*: raw web-scraped rental data lacks ground-truth labels for fraud. We address this by implementing a Programmatic Weak Supervision framework using the Snorkel library. This allows us to translate domain expertise into a series of Labeling Functions (LFs) that generate probabilistic labels for training discriminative models.

Our research methodology emphasizes three critical technical contributions: (1) Effective-Price Recovery via NLP heuristics, (2) Hyper-Local Market Baselines using the Zillow Observed Rent Index (ZORI) and spatial k-NN, and (3) a Comparative Supervision Analysis between generative weak supervision and hand-labeled ground truth.

1.1 Research Questions

This project aims to address the following central research questions:

1. To what extent can programmatic weak supervision, supported by external hyper-local market indices and NLP-based feature recovery, substitute for ground-truth labels in identifying rental fraud?
2. How do systematic biases in labeling functions (e.g., the assumption that low-price correlates with fraud) propagate through a Snorkel-based generative model and impact downstream discriminative performance?
3. Does a small, hand-labeled expert dataset outperform a massive but noisily-labeled dataset in the specific context of adversarial real estate fraud?

1.2 Target Audience

The results of this study are primarily intended for:

- **Data Scientists and Trust-and-Safety Engineers:** Professionals at online classified platforms who must handle high volumes of unlabelled data and require robust, scalable anomaly detection pipelines.
- **Computational Urban Researchers:** Scholars interested in the intersection of market economics and machine learning, specifically regarding how local rent indices, like ZORI can be used to normalize geographic features.
- **Consumer Protection Agencies:** Stakeholders who benefit from identifying the economic dissonance signals, such as price-to-market ratios, that most reliably indicate fraudulent intent.

2 Related Work

The detection of fraudulent activity in online classifieds is a specialized subset of anomaly detection that intersects machine learning, natural language processing, and computational urban economics. Unlike financial fraud, which relies on transactional time-series data, rental fraud is characterized by unstructured text, a lack of historical user reputation, and high geographic variance. This section provides a comprehensive review of the literature across five key pillars: traditional real estate anomaly detection, the paradigm of Data Programming, market-indexed feature engineering, adversarial NLP, and the challenges of class imbalance in weak supervision.

2.1 Multi-Modal Anomaly Detection in Real Estate

Traditional approaches to identifying rental fraud have historically focused on account-level metadata and simple heuristic filters. Early systems relied on identifying extreme price outliers or duplicate image detection across disparate platforms. A pivotal study by Park et al. [Park et al., 2017] characterized rental scams on Craigslist by combining listing text, contact behavior, and posting metadata, identifying cloned-listing patterns where photos from legitimate sales are repurposed in low-cost scams. However, as noted by recent investigations into platform-level security, modern aggregators (including those represented in the UCI Apartment Listings dataset [UCI Machine Learning Repository, 2019]) frequently strip image EXIF metadata to protect user privacy. This renders metadata-based cloning detection significantly less effective in the current ecosystem. Furthermore, linguistic models often struggle with the "marketing jargon" common to the real estate industry, leading to high false-positive rates for legitimate property managers. Our work addresses these gaps by shifting the focus from purely linguistic patterns to *economic dissonance*, the quantifiable gap between a listing's price and its verified geographic market value.

2.2 Data Programming and the Generative Labeling Paradigm

A persistent barrier in training robust fraud detectors is the labeling bottleneck or the Cold-Start Problem. In high-volume environments like the UCI collection (~99,000 observations), manual annotation is cost-prohibitive and suffers from high intra-annotator variance. The Snorkel framework, introduced by Ratner et al. [Ratner et al., 2017], revolutionized this space by formalizing Data Programming. In this paradigm, domain expertise is encoded into Labeling Functions (LFs). Unlike traditional hard-coded heuristics, Snorkel utilizes a generative model to estimate the latent accuracy of each LF by observing their matrix of agreements and disagreements across the unlabeled dataset. Mathematically, the framework assumes a latent variable model where the true labels are hidden, and the LFs serve as noisy observations. While Snorkel has seen massive success in medical imaging, its application to *adversarial domains* like fraud is technically precarious. As Varma et al. [Varma et al., 2019] argue, the dependency structure of LFs can lead to amplified biases if the domain assumptions are fundamentally misaligned with the dataset's hidden distribution. Our research specifically tests whether these generative models can discover fraud in a dataset where the base rate is extremely low and data has been heavily pre-moderated by upstream aggregators.

2.3 Market-Aware Feature Engineering and Spatial k-NN

The core of identifying a rental scam is determining if a price is too good to be true. However, fair market value is a moving target influenced by hyper-local factors. Traditional machine learning models often use global Z-score or Interquartile Range (IQR) filters. These are statistically flawed in a national dataset because a low price in Manhattan is a luxury price in a rural market. To solve this, we draw on Hedonic Pricing Models from urban economics, which suggest that a property's value is a function of its structural characteristics and its local environment.

Our work integrates the Zillow Observed Rent Index (ZORI) [Clark, 2022], which provides a smoothed measure of median rent. However, ZORI data is notoriously sparse at the ZIP-code level due to Zillow’s strict quality-check heuristics. To address this sparsity, we implement a spatial k-Nearest Neighbors (k-NN) fallback mechanism. This approach is grounded in *Tobler’s First Law of Geography*: "everything is related to everything else, but near things are more related than distant things." By calculating the median of the nearest ZORI-indexed neighbors within a 15km radius, we provide a geographically-aware market baseline for listings without direct ZIP matches. This represents a significant methodological refinement over city-wide averages, which often over-represent high-cost downtown cores at the expense of suburban or low-income neighborhoods.

2.4 Adversarial NLP and Effective-Price Recovery

A unique challenge in rental marketplaces is the deliberate use of placeholder pricing to manipulate search rankings. Many property management systems list buildings with a floor price of \$0 or \$1 to ensure they appear first when users sort by price. This is a form of *Adversarial Information Manipulation*, where the structured data field is intentionally inaccurate, while the "True" price is hidden in the natural language description (e.g., "*Units start at \$1,500*"). If left uncorrected, these listings act as noise that derails anomaly detectors, as a model may incorrectly identify a legitimate building advertisement as a high-probability scam. Our implementation utilizes custom regular-expression-based NLP heuristics to recover the "Effective Price" from the listing body. This technique mirrors Information Extraction (IE) tasks used in financial document processing but applies it to the rectification of systematic errors in real estate aggregator data. By comparing the structured price to the recovered effective price, we generate a high-signal feature that distinguishes between legitimate marketing spoofing and actual fraudulent intent.

2.5 Class Imbalance and Discriminative Bias

Finally, our work touches upon the literature regarding learning from imbalanced datasets. In fraud detection, the minority class (scams) often accounts for less than 1% of the total population. Standard discriminative models, such as Gradient Boosted Trees [Friedman, 2001], are prone to overfitting the majority class unless specific sampling or weighting techniques are employed. Our comparative analysis between Snorkel and hand-labeled supervised learning provides a case study in *bias propagation*. As noted in recent literature on weak supervision [Cachay et al., 2021], if weak labels are biased toward the majority class (legitimacy), the discriminative model will not only inherit this bias but may amplify it through its search for high-frequency feature interactions. Our work demonstrates that in extremely imbalanced and adversarial contexts, the wisdom of the crowd encoded in LFs can be systematically misled by deceptive but common listing patterns, necessitating a small but high-quality ground-truth evaluation set.

3 Methods

3.1 Datasets & Preprocessing

The primary dataset is the UCI Apartments for Rent Classified collection [UCI Machine Learning Repository, 2019], containing 99,492 rental listings from US platforms (91% from RentDigs.com, 7% from RentLingo). Each listing has 22 columns covering price, size, location, free-text body and title, amenities, source, and posting time, with no fraud labels.

The second dataset is the Zillow Observed Rent Index (ZORI) at the ZIP level [Zillow Research, 2026], a smoothed monthly mean of listed rents between the 35th and 65th percentile, published per region [Clark, 2022]. It spans 8,188 ZIPs over 135 months (Jan 2015 – Mar 2026)

and is intentionally sparse: Zillow only publishes the index for regions with sufficient data, leaving about 73% of ZIP-month cells empty.

Preprocessing was minimal. We drop the redundant `price_display` column, coerce `price` and `square_foot` to numeric, and remove 25 listings with missing or zero coordinates. We keep listings with other missing metadata, such as `address` (92% missing) and `pets_allowed` (61%), since those gaps may themselves carry signal. After cleanup, we have 99,467 listings.

3.2 EDA & Feature Engineering

Rent (\$1,350 median) and square footage (900 sqft median) are both right-skewed, listings cluster in major US metros, and missingness is highest in `address` (92%), `pets_allowed` (61%), and `amenities` (16%).

We engineer two within-listing anomaly signals. `price_per_sqft` normalizes price against unit size; a listing is a city-level price anomaly when its value falls outside $Q_1 - 1.5 \cdot \text{IQR}$ to $Q_3 + 1.5 \cdot \text{IQR}$ for cities with at least 20 listings (Figure 1). We also derive `amenity_count` and `pets_allowed_count` from the comma-separated fields to capture amenity-stuffing in bait listings.



Figure 1: Price versus sqft across all listings, with per-city IQR anomalies highlighted in red.

Since the UCI data lacks ZIP codes, we reverse-geocode each listing’s latitude/longitude to the nearest US ZIP centroid via `pgeocode` [Yurchak and Contributors, 2024] and a scikit-learn [Pedregosa et al., 2011] `BallTree` with the haversine metric. Direct (ZIP, month) lookups cover 47% of listings. City- or metro-level fallbacks would overestimate baselines for cheap neighborhoods inside expensive cities, so we instead impute missing baselines with same-county spatial k-NN, taking the median of up to five nearest ZORI ZIPs within 15 km. This adds 33% coverage at a median neighbor distance of 7.1 km. We track the resulting granularity (`zip`, `spatial`, or `none`) so downstream models can downweight imputed baselines.

Hand-labeling later revealed that aggregator listings often store the building’s floor price in `price` rather than the unit’s actual price, with the full range appearing in the body text (e.g. “*Monthly rental rates range from \$1,028 - \$1,608*”). To address this, we parse the body. If an explicit range (\$X to \$Y) is detected, we define the effective price as the upper bound (Y). Conversely, if only a floor is present (e.g., range from \$X), we apply a `has_unbounded_range` flag so downstream SCAM labeling functions can abstain on these listings.

From the matched baseline we derive five features: a `price_vs_zori_ratio`, a within-(ZIP,

month) `price_zori_zscore`, two asymmetric flags (`zori_anomaly_low` at $z < -2.0$ and `zori_anomaly_high` at $z > +2.5$, with global-percentile fallbacks for thin groups), and a continuous `zori_anomaly_strength` (absolute z-score).

3.3 Models

3.3.1 Model 1: Snorkel LabelModel

We define 17 labeling functions (LFs) across four families: ZORI- and price-based SCAM voters, hand-derived hunter heuristics for free-form luxury bait pricing, text and contact SCAM voters (phone, email, urgency keywords), and legitimate-explanation NOT_SCAM voters identifying student housing, Section 8 listings, aggregator templates, and explicit price ranges. Each LF emits SCAM, NOT_SCAM, or ABSTAIN per listing. We combine LF votes with Snorkel’s generative `LabelModel` [Ratner et al., 2017] (binary cardinality, 500 epochs, learning rate 0.01), which learns each LF’s accuracy from the agreement structure across LFs and outputs a probabilistic $P(\text{SCAM})$ per row. Our LF set is intentionally NOT_SCAM-heavy to counter the SCAM-skewed engineered features.

3.3.2 Model 2: Gradient Boosted Classifier on Weak Labels

The standard follow-up step in a Snorkel pipeline is to train a discriminative model on the `LabelModel`’s soft labels, with the goal that the model finds feature interactions the LFs missed. We choose gradient boosting [Friedman, 2001] because tree ensembles consistently outperform linear models on heterogeneous tabular data with mixed numeric and categorical features. We use `HistGradientBoostingClassifier` from `scikit-learn` with default hyperparameters and class weights to handle imbalance, trained only on listings where Snorkel is highly confident ($P(\text{SCAM}) > 0.8$ for positives and < 0.2 for negatives). Features include all engineered numeric and boolean signals plus one-hot encodings of `source`, `has_photo`, and `zori_granularity`.

3.3.3 Model 3: L1-Regularized Logistic Regression on Hand Labels

We additionally train a sparse logistic regression directly on our 165 hand-labeled rows (described in §3.4). With $n = 165$ and roughly 53 features, L1 regularization [Tibshirani, 1996] is the textbook choice: the lasso penalty drives uninformative coefficients to zero, producing an interpretable model that resists overfitting. The model minimizes

$$\hat{\beta} = \arg \min_{\beta} - \sum_{i=1}^n \log p_i(\beta) + \lambda \|\beta\|_1 \quad (1)$$

where $p_i(\beta)$ is the predicted scam probability for listing i and λ controls sparsity. We use `scikit-learn`’s `LogisticRegression` with the saga solver and default L1 strength. Predictions are generated via leave-one-out cross-validation, so each held-out row is predicted by a model trained on the other 164.

3.4 Training and Evaluation Methodology

Since the UCI dataset has no fraud labels, we constructed a held-out evaluation set by hand-labeling 165 listings across six rounds. Each round used stratified sampling targeting model blind spots: top- K by Snorkel probability, random over the abstain zone, hypothesis-driven pattern probes (no photo, contact info, urgency keywords), and a "hunter" stratum derived during labeling that surfaced scams at 6.7% versus 1.1% for random sampling. Each listing received one of `scam`, `not_scam`, or `unsure`. The hand-labeled set serves as a uniform held-out evaluation set for all three models, enabling direct head-to-head comparison. Models are trained as described in §3.3 on disjoint label sources: Snorkel learns from LF agreement structure, the GBM from

Snorkel’s confident soft labels, and the L1 logistic regression from the hand labels themselves. All three produce a scam probability per listing, which lets us compute AUROC, top- K precision, and the ROC curve on the same 165-row set, side by side. The Snorkel LabelModel and the GBM never see these labels. The L1 logistic regression uses them as training data but evaluates via leave-one-out cross-validation (LOOCV), so each prediction comes from a model that never saw the corresponding label.

We report AUROC for ranking quality on the strict subset (excluding unsmes) and top- K precision for triage relevance at $K \in \{10, 50\}$, treating unsmes as positives since a human reviewer would investigate them.

4 Results

4.1 Model Evaluation

We evaluate the three models against the 165 hand-labeled listings using AUROC (strict, computed on the 156 `scam/not_scam` cases only) and top- K precision (inclusive, with the 3 `unsure` cases counted as positives alongside the 6 confirmed scams). Table 1 and Figure 2 summarize the results.

Model	AUROC	AP	Top-10	Top-50
Snorkel LabelModel	0.45	0.04	0%	8%
GBM on weak labels	0.33	0.03	0%	8%
Hand-supervised LOOCV	0.82	0.21	30%	10%

Table 1: Performance on 165 hand-labeled listings. AUROC and AP exclude "unsure" labels, while Top- K treats them as positives.

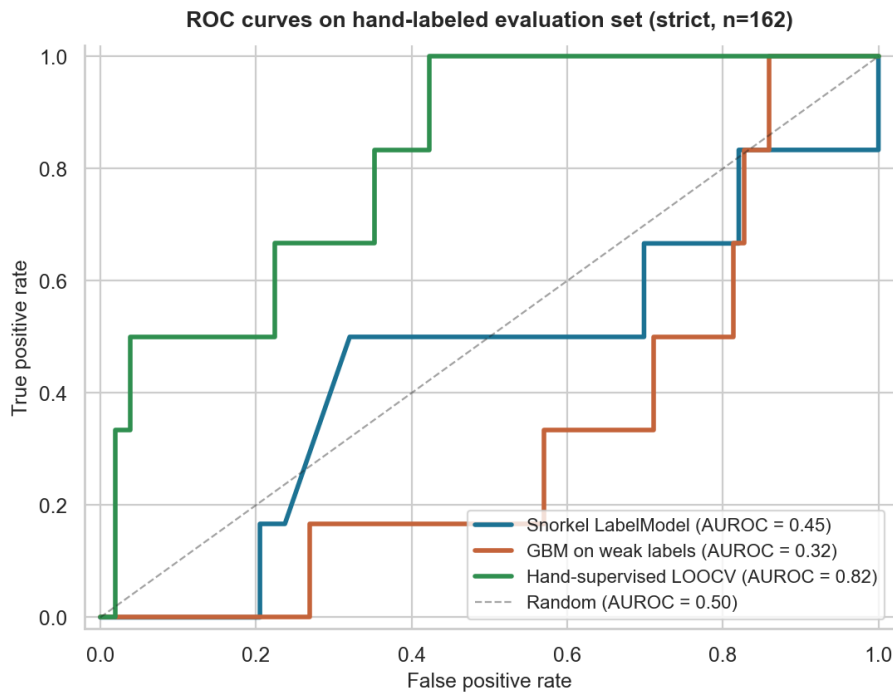


Figure 2: ROC curves for the three models on the 165 hand-labeled listings. The hand-supervised LOOCV model approaches the top-left corner, while the GBM curve sits below the diagonal.

The Snorkel LabelModel achieves AUROC 0.45 with top-50 precision of 8%, and zero of its top

10 highest-ranked listings are confirmed scams or unsure cases. The GBM trained on Snorkel’s confident soft labels reaches AUROC 0.33, the only model that performs worse than random ranking. Top-10 and top-50 precision are identical to Snorkel’s, suggesting the GBM inherited the LabelModel’s confident mistakes rather than learning to correct them. The hand-supervised L1 logistic regression trained on the 165 hand labels via LOOCV reaches AUROC 0.82 and top-10 precision of 30%, clearly separating real scams from non-scams.

4.2 Comparative Analysis

The hand-supervised model beats both weak-supervision approaches on every metric we report, despite being trained on roughly a hundredth of the data. This result contradicts the standard expectation that weak supervision combines many noisy signals into a stronger one. In our setting the noise was not symmetric. Our SCAM-leaning labeling functions encoded a domain assumption (cheap rent equals scam) that holds for unfiltered classified-ad data but fails on aggregator-cleaned datasets like ours, where most below-market listings are legitimate student housing, Section 8 units, or floor prices for multi-unit buildings. Because the assumption was systematically wrong, the LabelModel’s biased output carried through to the GBM, which learned to predict those mistakes with higher confidence. Only the L1 logistic regression, trained directly on the 165 hand-labeled rows, was protected from this bias.

5 Discussion

5.1 Connections to Prior Work

The Snorkel framework was originally proposed for settings where labeling functions have diverse, mostly uncorrelated noise [Ratner et al., 2017]. Our results suggest a sharper boundary: when LFs encode a systematically biased prior, the LabelModel’s output amplifies that bias rather than averaging it out, and downstream discriminative models trained on those weak labels do worse than the LabelModel itself. This is consistent with Snorkel’s reliance on LFs being better-than-random in expectation, but it shows how easily that assumption can break in practice when the domain prior is wrong about the underlying signal.

Our negative result on listing-text features echoes findings from production fraud detection. The Airbnb fraud system [Penchikala, 2018] is built primarily on account-level and behavioral signals, not listing content. The Dutch rental scam dataset [Serban, 2024] confirms this, with its most predictive features being advertiser completeness scores, login country mismatches, and profile picture presence, all platform metadata we did not have access to. Our work also shows the value of a hand-labeled evaluation set when applying weak supervision: without it, we would have reported Snorkel’s confident misclassifications as fraud detection.

5.2 Limitations

There are a few important limits on what we can claim. We cannot directly observe the true scam rate in our dataset, but random sampling of 90 listings during hand-labeling turned up one confirmed scam, suggesting a base rate close to 1% with wide uncertainty given the sample size. Either way, the rate is low enough to cap any classifier’s precision-recall trade-off and to make fine performance differences between models hard to evaluate. The dataset is dominated by aggregators (91% RentDigs.com, 7% RentLingo) that filter obvious scams upstream, so the listings that reach us are systematically less fraudulent than what a renter would actually encounter on a less-moderated platform. We also lack the platform metadata (account age, login country, profile completeness) that production systems treat as primary signals, leaving listing text and rent levels to substitute for signals they cannot reliably provide. Finally, our hand-labeled evaluation

set is small ($n = 165$) and stratified rather than random, so the AUROC of 0.82 we report for the L1 logistic regression reflects performance on the harder, more anomalous slice of the data, not a random listing. The relative ordering of the three models would likely hold on a larger random sample, but the absolute numbers may not.

6 Conclusion & Future Work

We set out to build a fraud detector for rental listings using only listing content and a public market-rent baseline. Our hand-supervised L1 logistic regression on 165 hand-labeled rows reached AUROC 0.82 and top-10 precision 30%, while Snorkel’s LabelModel and a downstream GBM trained on its confident labels reached AUROC 0.45 and 0.33. The key takeaway is that biased labeling functions can make weak supervision worse than no model at all: when LFs encode the wrong domain assumption, the LabelModel propagates those mistakes downstream, and the discriminative model trained on its outputs performed worse than random on real fraud.

Beyond this finding, the paper contributes two reusable methodological contributions. The first is a spatial k-NN imputation scheme for Zillow ZORI baselines that adds 33% coverage while avoiding the bias of city- and metro-level aggregation. The second is a hand-labeling protocol with hunter strata that found scams at six times the random rate and produced the most predictive single LF in our pipeline.

Future work should focus on the features production fraud systems actually use: account age, login country, profile completeness, image features, and active probing of inquiries. A complementary direction is to scrape a less filtered source such as Craigslist or unfiltered Zillow, where the underlying scam population is denser. A larger hand-labeled set would also help close the gap between the hand-supervised model and a properly tuned discriminative baseline.

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